

Mooring Design Study

Soft moorings for the single buoy, the 4-buoy cluster and the 4×4 platform: criteria, selection, verification and installation checks

Specification complete; interference and excursion met

Status Ready for procurement Rev C.1 is complete. Procurement is gated only on HWRL confirming facility assumptions F1–F7, e.g. underwater wall anchors rated ≥ 300 N.	Criterion 6 Interference within $\zeta/3$ Tilt natural period shifts $-1.37 / -0.73 / -0.72$ % (buoy / cluster / platform) against $\zeta/3 = 1.42 / 1.19 / 1.05$ %. Heave natural period shifts ≤ 0.50 %.	Criterion 4 Excursion ≤ 1.0 m At $H = 0.5$ m the extreme cord set (k $\times 5 / \times 6$) holds the max surge to $0.97 / 0.95$ m. The operational set would drift to $3.39 / 3.53$ m.
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One anchor layout, two cord sets: **operational** ($H \leq 0.12$ m, minimal interference) and **extreme** ($H = 0.2\text{--}0.5$ m, station-keeping).

Three articles, natural periods inside the wave band

Article	Configuration	Mooring points	Heave T _n	Tilt T _n
Single buoy	1 spar with a heave plate	radial collar, 4 lines	2.561 s	2.761 s
Cluster	4 buoys hinged to a hub, at 45°	4 spars, 4 lines	2.586 s	2.856 s
4×4 platform	16 buoys at 45°, 2.06 m wide	8 row spars, 4 bridles	2.606 s	2.924 s

Band	Wave height H	Wave period T (regular, heading 0)	Purpose
Operational	0.04, 0.08, 0.12 m	1.4–3.5 s; 0.05 s step across the resonances	motions: FloatSim validation
Extreme	0.2, 0.35, 0.5 m	a subset, steepness $H/\lambda \leq 0.08$	line and anchor loads, offsets

Mesh correction: these T_n are long

T_n: unmoored, FloatSim modal analysis on the flume BEM. Its 12-sided waterline reads C33 and C55 4.5 % low, so every T_n above is long by ~0.04 s (heave) and ~0.06–0.08 s (tilt). Converged mesh, single buoy: **heave 2.519 s, pitch 2.684 s**, matching the OSU buoy model (2.52 / 2.69 s) and the field heave-decay video (≈ 2.50 s). The mooring's shifts are relative and unaffected.

Station-keeping and non-interference conflict

Station-keeping

- max surge, mean + dynamic, ≤ 1.0 m: tracking window $-0.3 \dots +1.0$ m
- $|\text{mean sway}| \leq 0.10$ m; wall clearance ≥ 0.6 m
- no slack in the operational band: $T_{\min} \geq 0.15 T_0$
- $WLL \geq 3 \times \text{max tension}$; stroke $\geq 1.25 \times \text{max stretch}$

Non-interference

- tilt natural period shift $\leq \zeta/3$: 1.42 / 1.19 / 1.05 %
- heave natural period shift ≤ 1 %
- calm static tilt $\leq 1^\circ$
- surge, sway, yaw periods $\geq 4 \times 3.5 \text{ s} = 14 \text{ s}$

Stiff lines hold station; soft lines minimise interference. $\delta \leq \zeta/3$ keeps the resonant amplitude $\geq 1/\sqrt{(1 + 1/9)} = 94.9$ % of the unmoored peak.

Criteria first, every option simulated in FloatSim

- 1 Criteria agreed before design.** Seven pass/fail criteria, confirmed in Phase B.
- 2 FloatSim for every option.** Moored decks, static settle, modal analysis, rigid pulls, and regular-wave runs with wave-relative Morison drag and the drift sum applied in-run.
- 3 Selection by a written rule.** Hard criteria, then the most pretension, the attachment closest to the waterline, the stiffest surge.
- 4 Verification runs.** The chosen designs in FloatSim wave runs, operational band and $H = 0.5$ m.
- 5 Generated documents.** The specification's tables and figures are built from the same JSON records.

Drift sum: the recorded fixed-body bound (splash-zone drag + Havelock term) plus FloatSim's own mean force, so conservative. LEVEL1 kinematics are valid to 5.7° : resonant results are indicative.

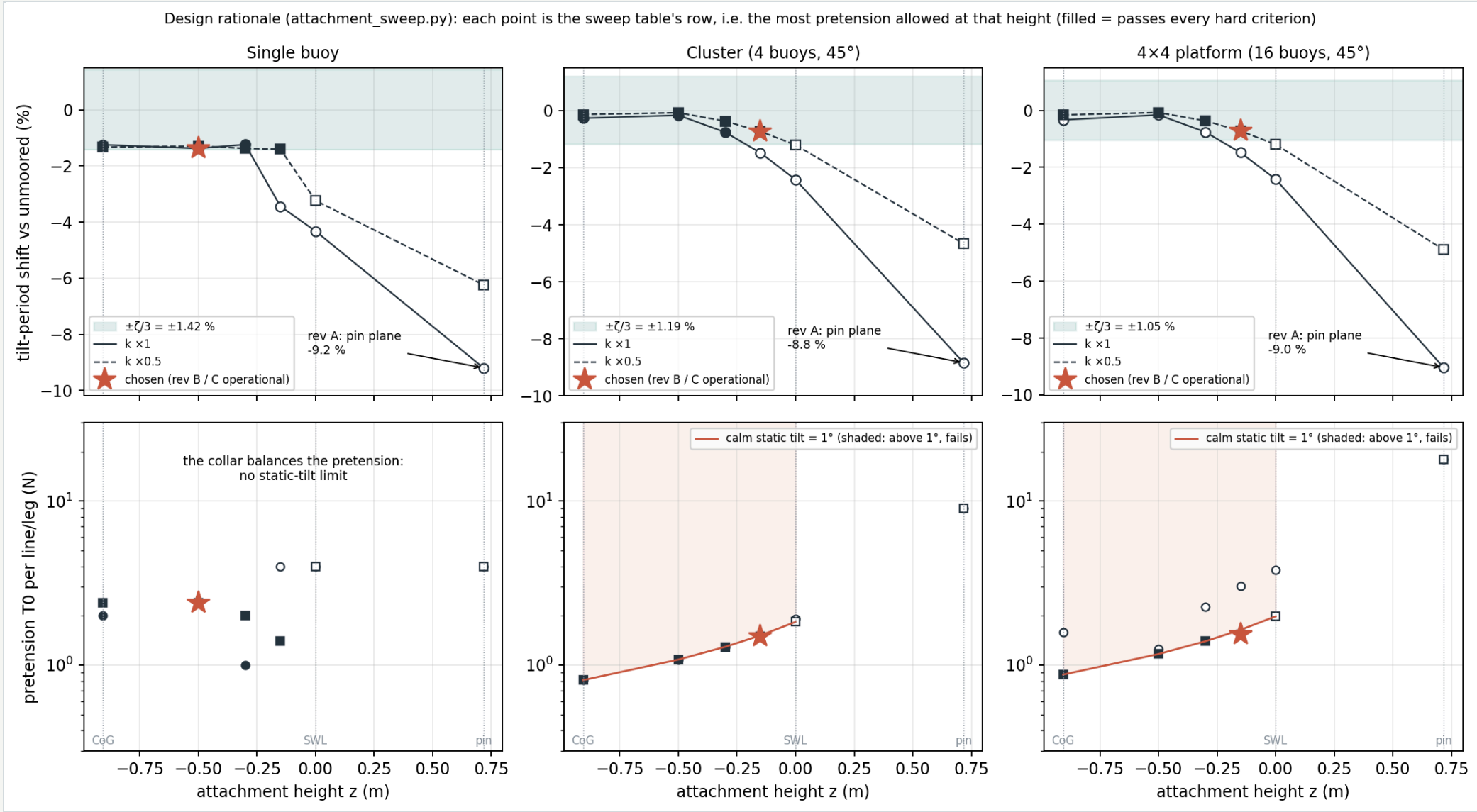
Four revisions, each correcting the last

Rev	Design	Finding
A	Lines at the pin plane (+0.717 m)	Tilt T_n -9.19 / -8.83 / -9.04 %, 6.5–8.6 × ζ/3: withdrawn
B	Attachment sweep: z -0.50 / -0.15 / -0.15 m, low T0	Within ζ/3, but H = 0.5 m drifts 2.5–3.4 m: criterion 4 was missing from the rule
C	Rev B as the operational set + an extreme set, k ×5 / ×6, same anchors	Max surge 0.97 / 0.95 m at H = 0.5 m; its surge periods 10.5 / 9.6 s declared
C.1	Creep allowance, ~4 % of the stretch per decade (assumed)	Creep table, daily re-check and re-tension thresholds

Phase H evaluated removing the pretension and kept it (no spec change): see the decisions.

Key insight • attachment height

Attach below the waterline: coupling $\approx k \cdot h^2$



Two reference points

The static pretension moment acts about the pin (+0.717 m). The dynamic coupling, $\approx k \cdot h^2$, acts about the tilt mode's rotation centre, near the CoG (-0.907 m).

Tilt T_n shift, buoy / cluster / platform

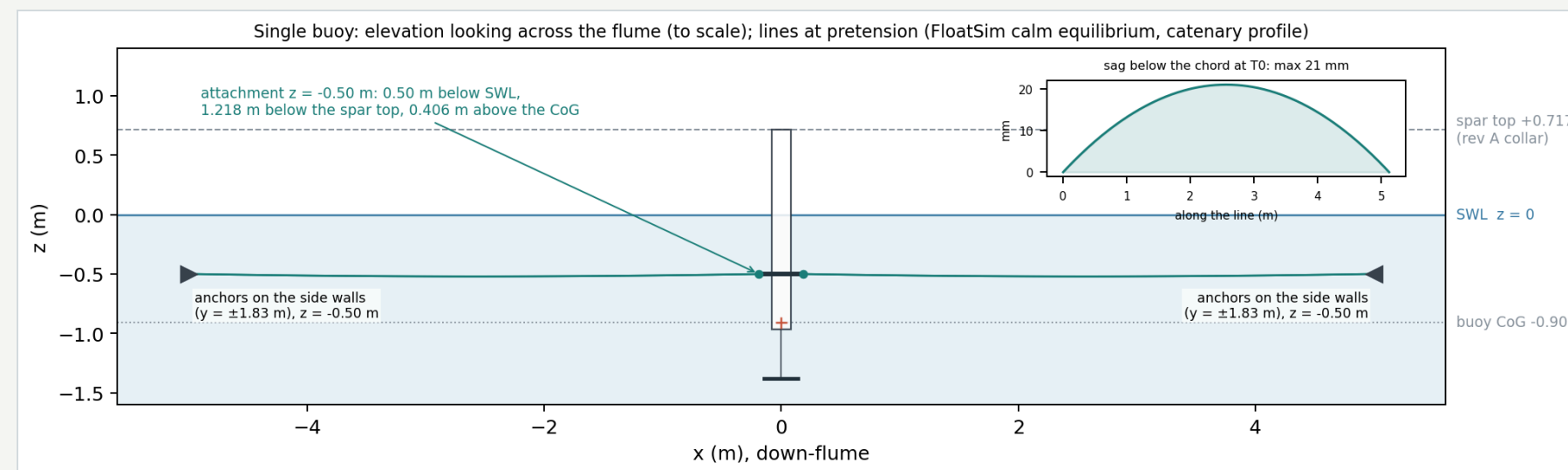
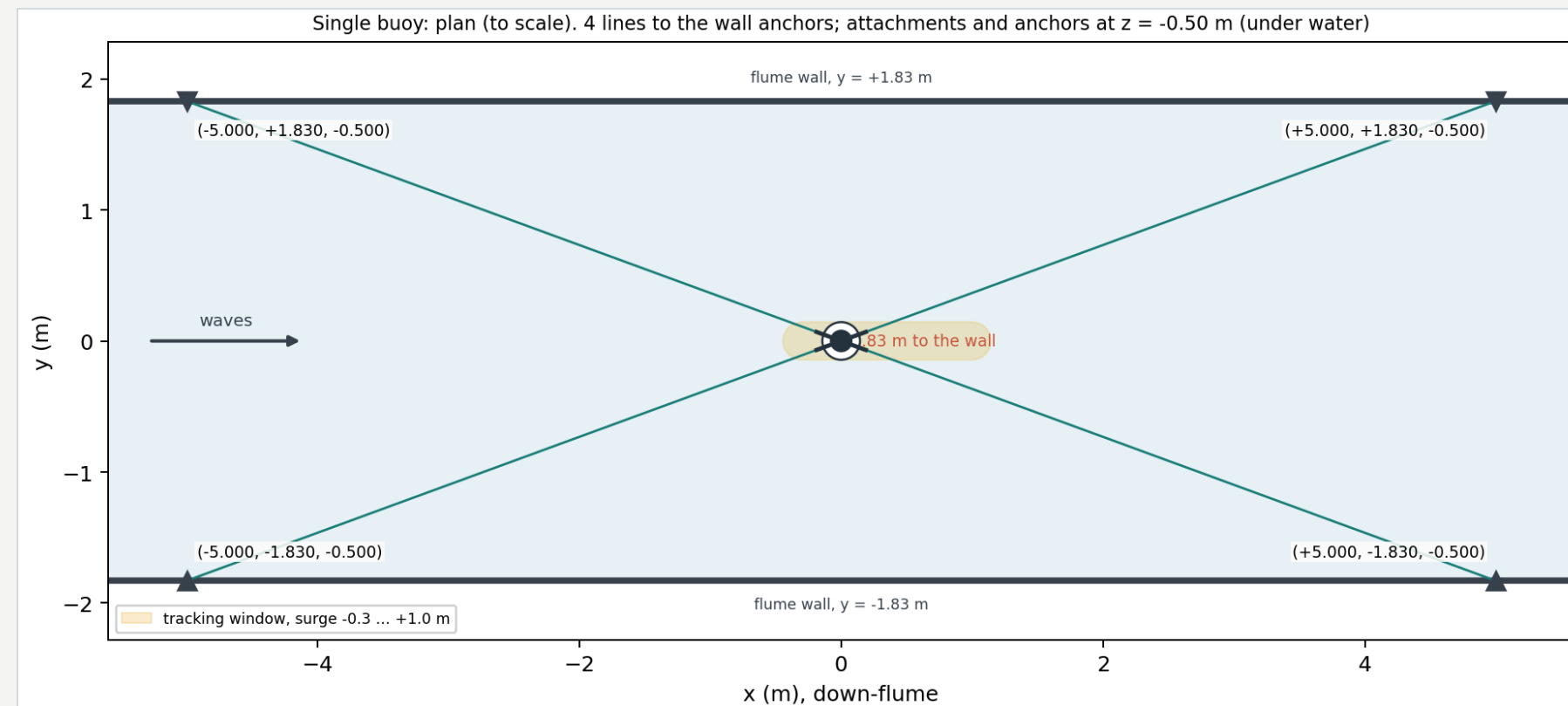
Pin plane: -9.2 / -8.8 / -9.0 %

Chosen: -1.37 / -0.73 / -0.72 %

Limit $\zeta/3$: 1.42 / 1.19 / 1.05 %

Trade-off

Below the pin, the pretension tilts the cluster and platform: 1.00° calm tilt, $\approx 0.70^\circ/\text{N}$, used as the pretension gauge.



Configuration 1 of 3 · to scale

Single buoy

Four lines from a radial collar to wall anchors at $x = \pm 5.0$ m, $y = \pm 1.83$ m. The collar balances the lines: calm tilt 0° .

Attachment

radial collar r 0.2 m, z -0.50 m

Cord, one set

k 2.01 N/m, T_0 2.40 N at rest

Heave T_n , free $\rightarrow$ moored (flume BEM)

2.561 $\rightarrow$ 2.548 s (-0.50%)

Tilt T_n , free $\rightarrow$ moored (flume BEM)

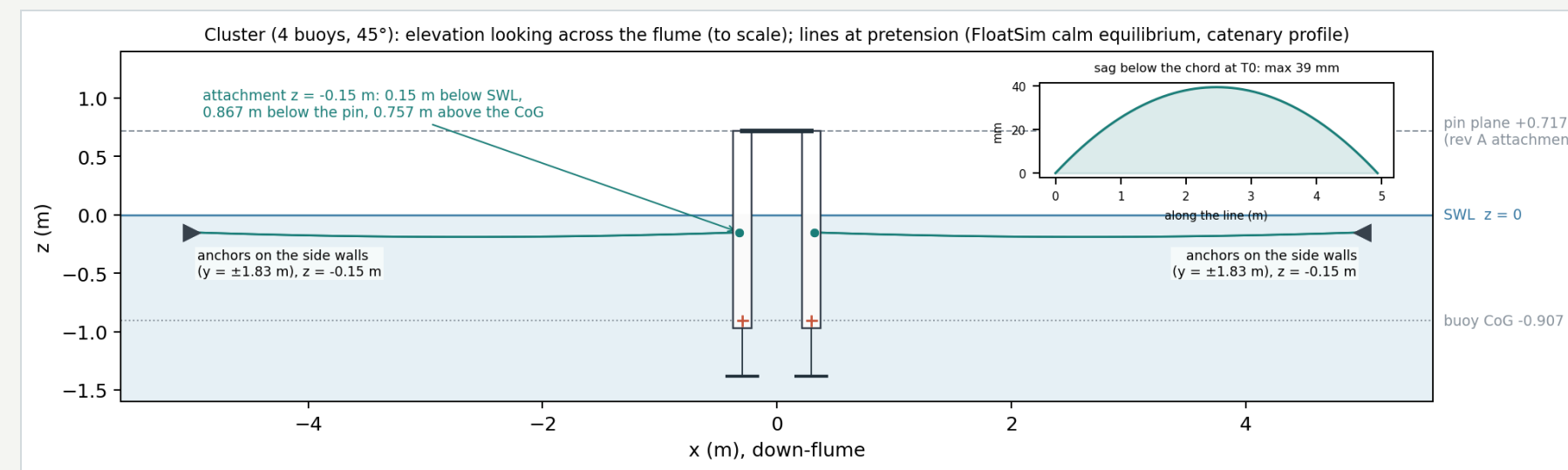
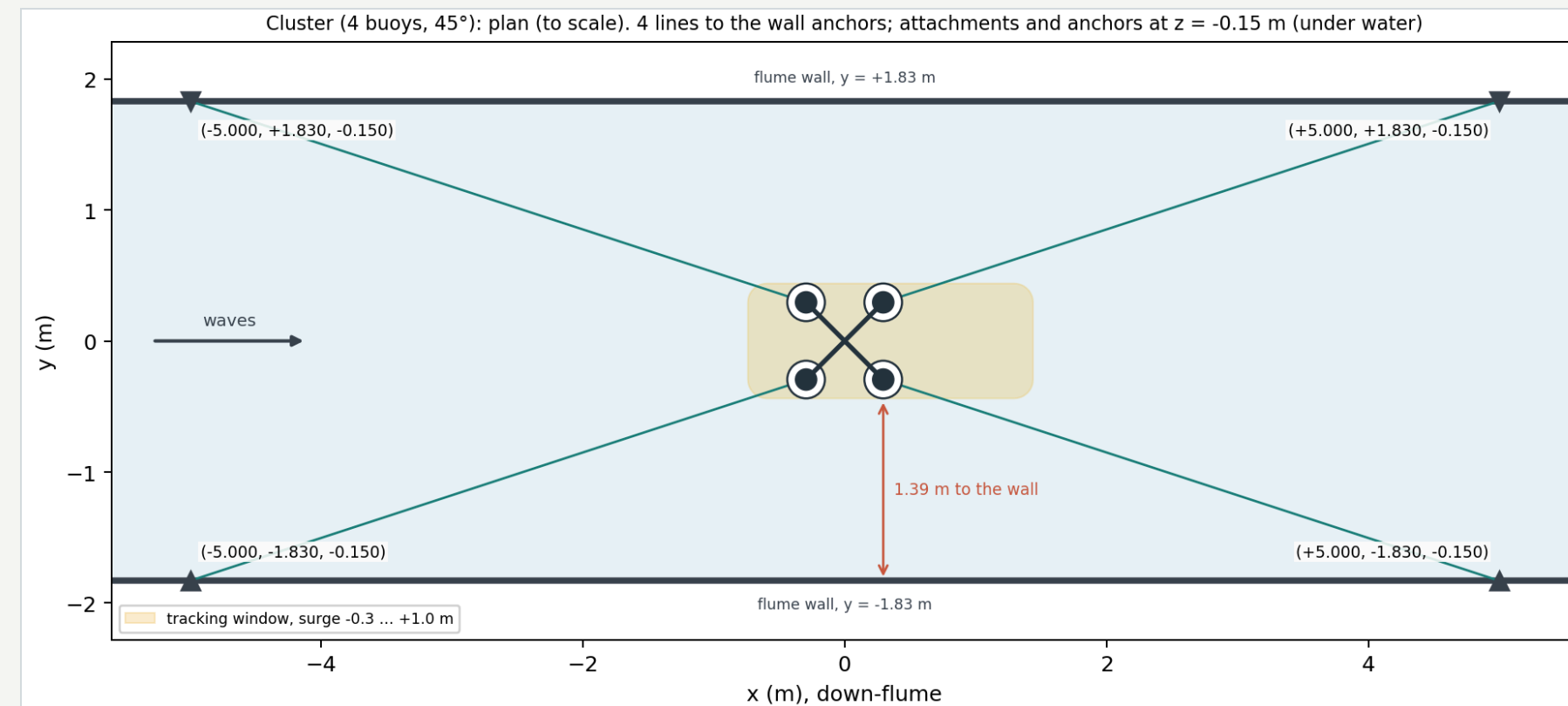
2.761 $\rightarrow$ 2.723 s (-1.37%)

Converged mesh, free: heave · pitch

2.519 s · 2.684 s (video $\approx$ 2.50 s)

$H = 0.5$ m max surge

0.68 m (predicted cases)



Configuration 2 of 3 · to scale

Cluster

4 buoys hinged to a hub at 45°. One line per spar to wall anchors at $x = \pm 5.0$ m, $y = \pm 1.83$ m.

Attachment

each spar, $z -0.15$ m

Cords, operational / extreme

k 4.15 / ≥ 20.75 N/m, 1.44 N at rest

Heave T_n , free $\rightarrow$ moored

2.586 $\rightarrow$ 2.584 s (-0.08%)

Tilt T_n , free $\rightarrow$ moored

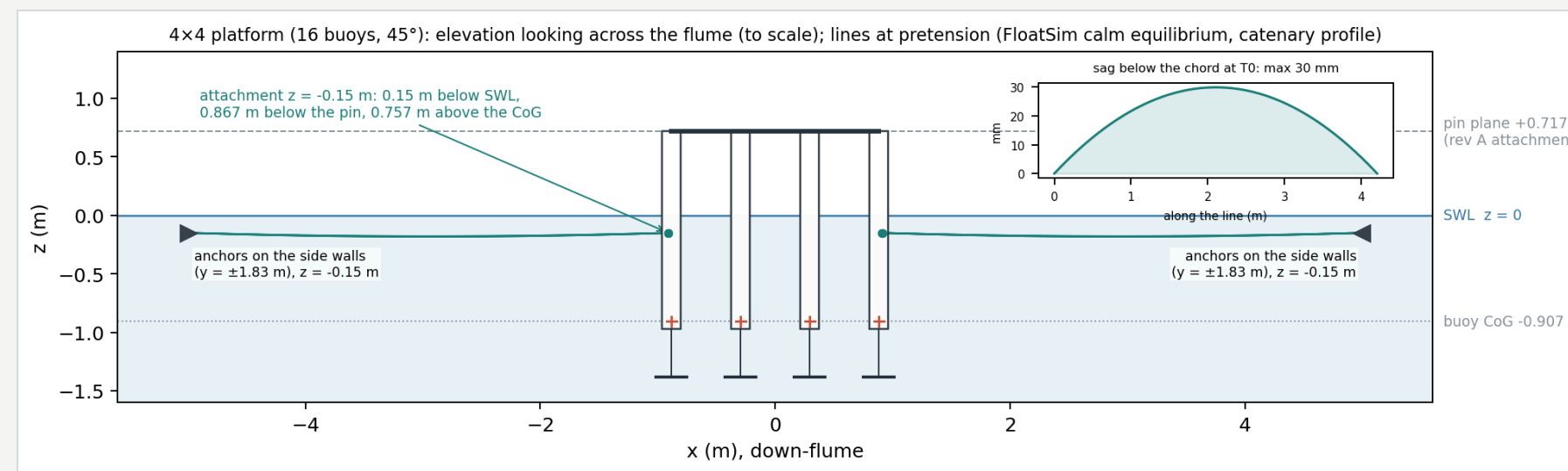
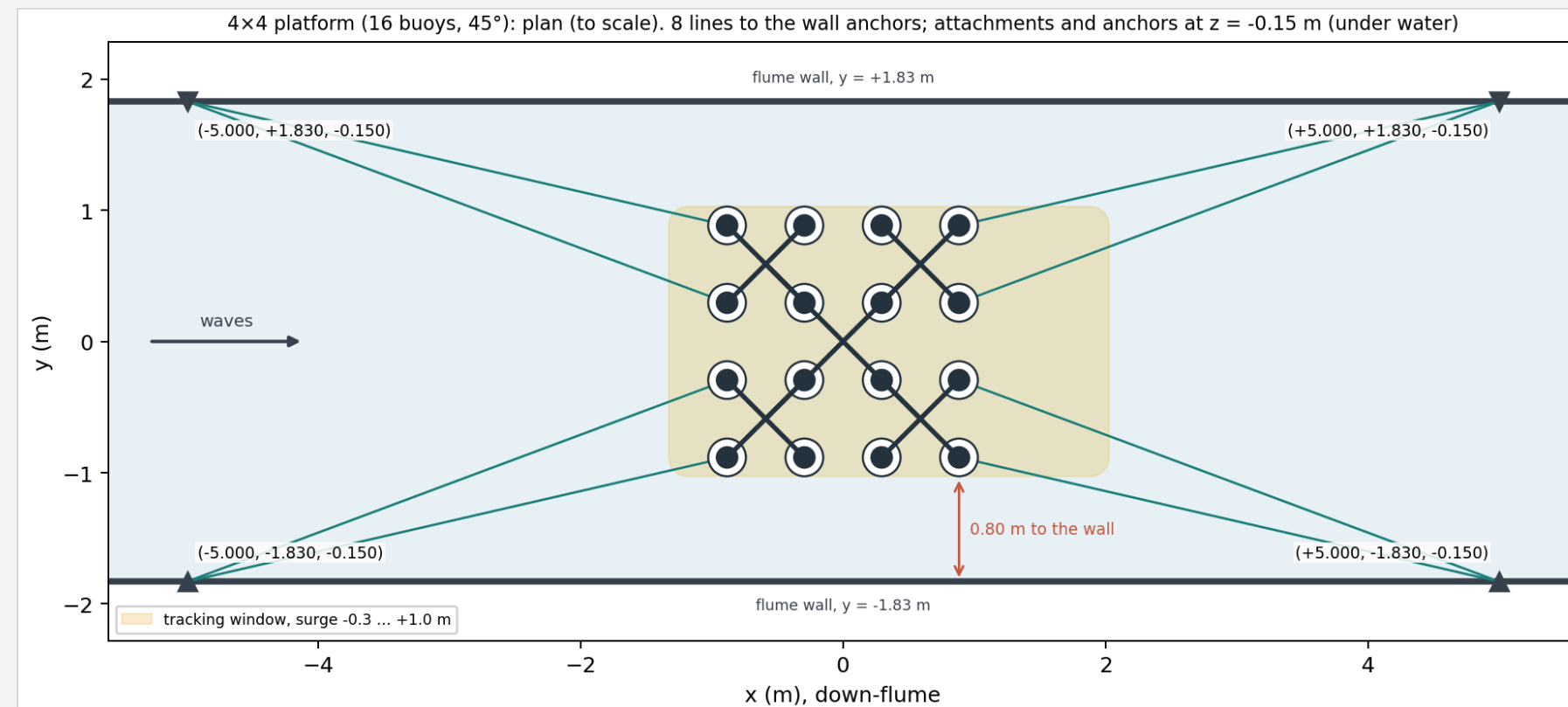
2.856 $\rightarrow$ 2.835 s (-0.73%)

Calm tilt · wall clearance

1.00°, both sets · 1.39 m

$H = 0.5$ m max surge, extreme set

0.97 m



Configuration 3 of 3 · to scale

4x4 platform

16 buoys at 45°, 2.06 m wide. Four 2-leg bridles to the row spars; anchors at $x = \pm 5.0$ m, $y = \pm 1.83$ m.

Attachment

8 row spars, $z = -0.15$ m

Cords per leg, operational / extreme

k 8.32 / ≥ 49.91 N/m, 1.42 N at rest

Heave T_n , free $\rightarrow$ moored

2.606 $\rightarrow$ 2.606 s (−0.03 %)

Tilt T_n , free $\rightarrow$ moored

2.924 $\rightarrow$ 2.902 s (−0.72 %)

Calm tilt · wall clearance

1.00°, both sets · 0.80 m

$H = 0.5$ m max surge, extreme set

0.95 m

The mooring barely moves the natural periods

Article	Tilt T _n shift (limit ζ/3)	Heave T _n shift (limit 1 %)	Heave RAO Δ vs free at the tilt resonance, H 0.04 / 0.12 m	Calm tilt
Single buoy	−1.37 % (1.42)	−0.50 %	−0.2 / +2.5 %	0°
Cluster	−0.73 % (1.19)	−0.08 %	+3.2 / +1.9 %	1.00°
4×4 platform	−0.72 % (1.05)	−0.03 %	+4.6 / +2.8 %	1.00°

Heave

The natural period moves ≤ 0.50 %, about 1/7 of the heave ζ/3 (ζ ≈ 11 %). The largest heave RAO change, +9.0 % (buoy) and +7.8 % (cluster) at H = 0.12 m, is at T = 2.2 s: off resonance, inside the slosh exclusion.

Tilt

Peak tilt, moored – free: +0.97 to +1.08° off resonance, i.e. the 1° calm tilt; +0.45° at the cluster and platform resonance; buoy ≤ +0.39°.

Open: Phase D

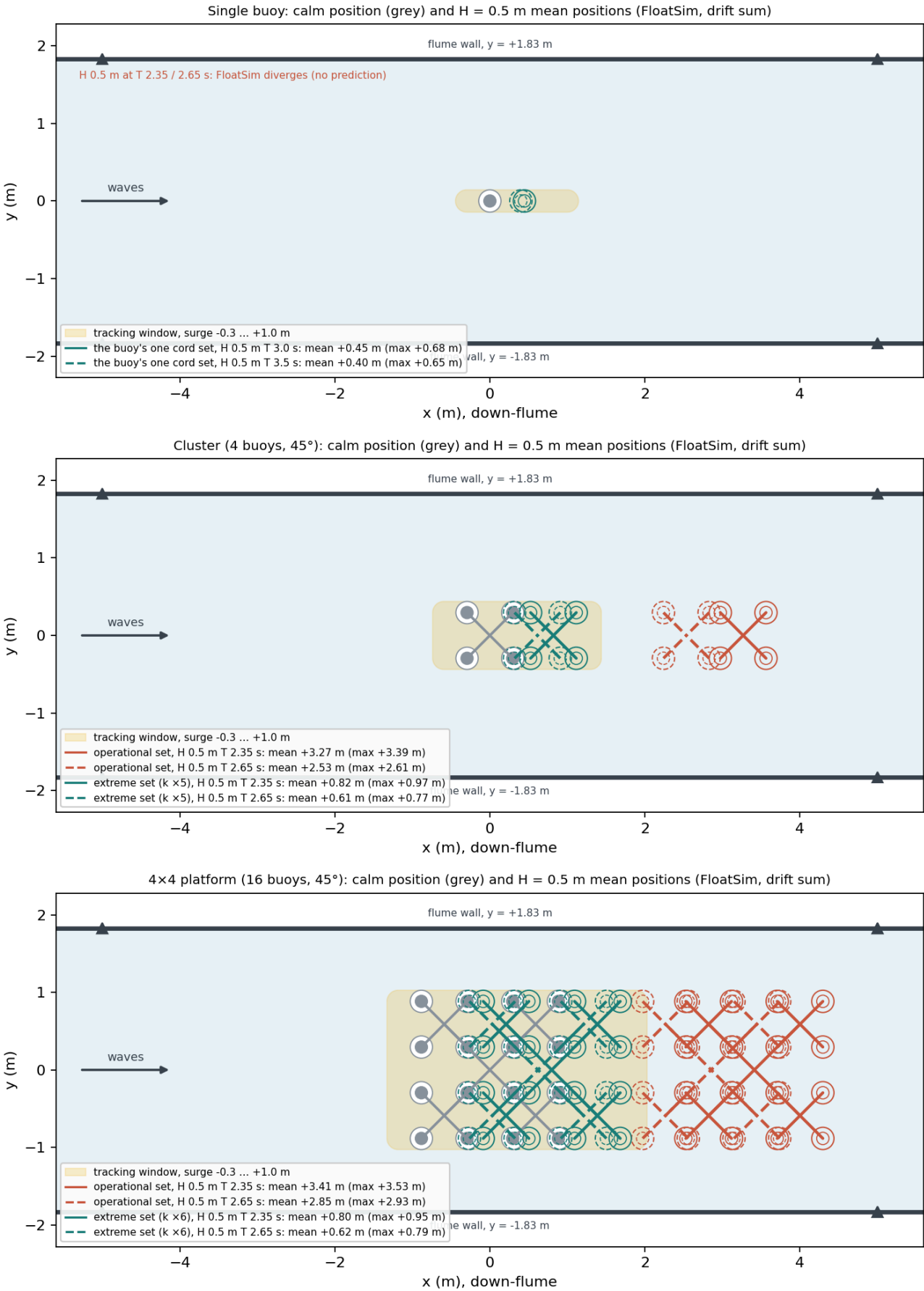
Criterion 6 asks ±3 % on the heave RAO; +3.2 / +4.6 % and the 2.2 s values exceed it. The formal check, with restrained free runs, is Phase D.

Result 2 · excursion at H = 0.5 m, FloatSim, drift sum

The extreme set keeps H = 0.5 m in view

Article	Cord set	T 2.35 s: mean / max	T 2.65 s: mean / max
Cluster	operational	3.27 / 3.39 m	2.53 / 2.61 m
Cluster	extreme ×5	0.82 / 0.97 m	0.61 / 0.77 m
Platform	operational	3.41 / 3.53 m	2.85 / 2.93 m
Platform	extreme ×6	0.80 / 0.95 m	0.62 / 0.79 m

- Operational band ($H \leq 0.12$ m): mean offset ≤ 0.07 m, no slack ($T_{\min} / T_0 \geq 0.19$).
- At $H = 0.5$ m the down-flume lines go slack and stay slack; the up-flume lines carry the drift (peak 22.1 / 45.3 N).
- Single buoy: one set; its predicted $H = 0.5$ m cases reach 0.68 m.



One anchor layout, two cord sets

Cluster / platform	Operational set	Extreme set
Use	response tests, $H \leq 0.12$ m	loads and survival, $H = 0.2\text{--}0.5$ m
Cord stiffness k	4.15 / 8.32 N/m (−13 / +30 %)	20.75 / 49.91 N/m (−0 / +30 %)
At-rest tension · calm tilt	1.44 / 1.42 N · 1.00°	the same (at-rest-matched) · 1.00°
Pre-stretch at rest	346 / 171 mm	69 / 28 mm: set by tension, not length
Surge natural period	23.1 / 23.0 s	10.5 / 9.6 s (criterion 2 declared)
Surge pull stiffness K	15.0 / 60.6 N/m	73.2 / 349.3 N/m

Identification: the calm tilt cannot tell the sets apart, so a static pull is required before every extreme series: the extreme set reads $K \geq 33$ N/m (cluster) or ≥ 146 N/m (platform).

Key decisions and their technical basis

Decision	Basis
Keep the operational pretension	Taut lines in every operational case. Without it, the lines lose stretch for 18–40 % of each resonant cycle, which the quasi-static line cannot model.
Keep the buoy's pretension	The collar turns T0 into yaw stiffness; at the taut minimum the yaw period would be 2.48 s, inside the wave band.
Two cord sets, not one compromise	Criteria 2 and 4 cannot both hold at H = 0.5 m: surge ≥ 14 s caps k at $\sim \times 2.8$, leaving $\gtrsim 1.5$ m of offset.
Size the extreme set on max surge	Criterion 4 is mean + dynamic ≤ 1.0 m; the mean alone would allow $\times 4.5$ / $\times 5$.
Match the at-rest tension	Identical calm line forces: static residual 0.0073 / 0.0022 N, against 0.13 / 0.24 N at the same nominal T0.

Anchors, cords, collar and the daily re-check

Wall anchors

Four per article at the attachment depth, under water, each rated ≥ 300 N. The largest working load ($\times 3$ on the FloatSim peak) is 267.4 N: platform, up-flume, extreme set.

Elastic cords

Usable submerged, ≈ 0.02 N/m. Operational stiffness $-13 / +30$ %; extreme stiffness a minimum, $-0 / +30$ %. Elongation capacity $\times 1.25$ on the peak stretch.

Buoy collar

A slender spreader, round section ≤ 25 mm (added-mass budget 0.4 kg). A solid disk of $r = 0.2$ m would add ≈ 21 kg of heave added mass: rejected.

Creep: daily re-check

~ 4 % per decade assumed: 87.4 % of the at-rest tension is left after a day (calm tilt $1.00 \rightarrow 0.87^\circ$). Re-tension below the thresholds; the platform (operational) and the buoy before their critical cases.

Risks and limitations

Item	Handling
HWRL facility assumptions F1–F7 unconfirmed	Confirm before procurement; any “no” forces a redesign
LEVEL1 valid to 5.7°; resonant tilts 10–33°	Resonant results indicative; LEVEL2 rerun of the resonant subset
Buoy extremes: 3 cases diverge, 6 grow yaw	Run last with a height ramp, stop on yaw growth; pitch-neutral collar
Heave RAO change above ±3 % in some cases	Formal criterion-6 check with restrained free runs in Phase D
Snap loads and the return swing not simulated	×3 load factor; load-cell pull test at installation
Flume BEM waterline 12-sided (NT12)	Resonances ~0.04–0.08 s long; NT36 regeneration pending

Next steps

From specification to the flume

- 1** **HWRL confirmation of F1–F7.** Underwater wall anchors, the ≥ 300 N rating, cord in two stiffnesses: gates procurement.
- 2** **Procurement.** Both cord sets to the line tables; creep rate from the datasheet or a one-day hold test.
- 3** **Verification.** BEM at NT36 and the Phase D matrix: 442 cases, both cord sets, criterion 6 included.
- 4** **LEVEL2.** Rerun of the resonant subset: tilts, loads, offsets and the buoy's divergence.

Where every number comes from

Record	Holds
MOORING-SPEC.md / .pdf, rev C.1	the specification: line tables, figures, checks, procedure
DESIGN-BASIS.md	every decision and its evidence, Phases A–H
attachment_sweep.py → .json	the operational set: the sweep, the rule, the natural periods
extreme_set.py → .json	the extreme set: k scan, final runs, residuals
spec_statics.json, spec_statics_extreme.json	statics: anchors, attachments, T0, pull K, periods
pretension_study.json, PRETENSION-EVAL.md	the pretension evaluation
viewer_rows_revc/ (48 runs)	moored vs free comparison; the 3D motion viewer
mooring_figures.py → figs/	the plan, elevation, rationale and offsets figures